

Cone-Rod Dystrophy and Hearing Loss (CRDHL): A Comprehensive Disease Profile

Disease: Cone-Rod Dystrophy and Hearing Loss (CRDHL) **MONDO ID:** MONDO:0014980 · **OMIM:** #617236 · **Category:** Genetic (autosomal recessive ciliopathy) **Primary gene:** *CEP78* (HGNC:26820) · **Second locus:** *CEP250* (C-Nap1)

Evidence base: human clinical case series/cohorts, cell-biology (in vitro), and one knockout-mouse model. This is an ultra-rare disorder; several sections rely on a small number of primary reports plus analogy to related deaf-blindness syndromes, which is flagged where used.

Summary

Cone-Rod Dystrophy and Hearing Loss (CRDHL) is a rare, autosomal recessive **ciliopathy** whose principal molecular cause is **biallelic loss-of-function of *CEP78***, a centriolar distal-end protein that governs centriole length and centrosome homeostasis. First delineated as a distinct Mendelian entity by Namburi et al. in 2016, CRDHL couples a **childhood/adolescent-onset, cone-predominant cone-rod dystrophy** (leading to legal blindness by mid-adulthood) with **bilateral, post-lingual, progressive sensorineural hearing loss (SNHL)**, and — in a subset of male patients — **reduced fertility due to oligoasthenoteratozoospermia**. This triad is mechanistically unified by a defect in the specialized cilia and centriole-derived structures of the body's most centriole-dependent post-mitotic cells: photoreceptor connecting cilia, cochlear hair-cell kinocilia/basal bodies, and sperm flagella.

CRDHL is genetically and clinically distinguishable from **Usher syndrome** (the most common cause of combined deaf-blindness), which is typically rod-first with congenital hearing loss. A second locus, ***CEP250/C-Nap1***, produces an overlapping but milder CRD+SNHL phenotype and is otherwise classically associated with "atypical Usher syndrome." CRDHL therefore sits within a genetically heterogeneous group of retinal-dystrophy-plus-hearing-loss syndromes whose differential diagnosis includes *ARSG* (Usher type IV), *ABHD12* (PHARC), *TUBB4B*-related syndromic Leber congenital amaurosis, and *CEP250*.

There is **no disease-specific therapy**. Management is supportive and rehabilitative: low-vision aids, hearing aids, and — for severe-to-profound SNHL — **cochlear implantation**, which restores speech perception effectively in analogous post-lingual deaf-blindness syndromes. Diagnosis integrates multimodal ophthalmic phenotyping (full-field ERG showing cone-worse-than-rod loss, OCT, fundus autofluorescence), pure-tone audiometry, and molecular confirmation by retinal-dystrophy gene panels or whole-exome/whole-genome sequencing. This report synthesizes 12 confirmed findings drawn from 35 reviewed papers across human clinical, model organism, and in vitro evidence.

Key Findings

1. CEP78 biallelic loss-of-function causes CRDHL, distinct from Usher syndrome (F001)

The founding evidence for CRDHL as a *CEP78*-related disorder comes from **Namburi et al. 2016 (American Journal of Human Genetics)**. Using homozygosity mapping followed by whole-exome sequencing and founder-mutation screening, they identified **two truncating variants in CEP78** — c.893-1G>A (a splice-acceptor variant) and c.534delT — in **six individuals of Jewish ancestry** with autosomal recessive cone-rod degeneration and SNHL. The c.893-1G>A variant causes skipping of exon 7 (a 65-bp deletion), producing a frameshift **p.Asp298Valfs*17**. Crucially, immunohistochemistry of human retina showed **intense CEP78 labeling of cone inner segments relative to rods**, providing an anatomical basis for the cone-predominant degeneration.

"Homozygosity mapping followed by whole-exome sequencing (WES) and founder mutation screening revealed two truncating rare variants (c.893-1G>A and c.534delT) in CEP78, which encodes centrosomal protein 78, in six individuals of Jewish ancestry with CRD and SNHL."
— PMID: 27588452

This finding establishes *CEP78* truncating variants as the cause of a **new, non-Usher deaf-blindness syndrome**, distinguished by its **cone-first** (rather than rod-first) retinal degeneration and **post-lingual** (rather than congenital) hearing loss.

2. CEP78 is a centriolar distal-end protein controlling centriole length and centrosome homeostasis (F002)

The molecular function of CEP78 places CRDHL firmly among the **ciliopathies**. **Hossain et al. 2017 (Journal of Cell Biology)** showed that CEP78 localizes to **mature centrioles**, directly interacts with **VprBP (viral protein R binding protein)**, and **inhibits the EDD-DYRK2-DDB1 E3 ubiquitin ligase** to regulate centrosome homeostasis.

"Here, we show that centrosomal protein of 78 kDa (Cep78) localizes to mature centrioles and directly interacts with viral protein R binding protein (VprBP)." — PMID: 28242748

Karasu et al. 2022 further demonstrated that the **CEP350-FOP scaffold recruits CEP78 (or OFD1) to the distal end of centrioles**, and that the CEP350-FOP/CEP78 complex **controls centriole microtubule length**:

"recruiting the proteins CEP78 and OFD1 to the distal end of centrioles and promoting the assembly of subdistal appendages" — PMID: 36315013

Consistent with this, **Ascari et al. 2020** found that **patient-derived fibroblasts and nasal brushings show elongated primary cilia**, indicating impaired cilia assembly downstream of CEP78 loss:

"Elongated primary cilia without clear ultrastructural abnormalities in sperm or nasal brushes suggest impaired cilia assembly." — PMID: 31999394

3. The phenotype spectrum includes male infertility; a missense founder allele expands variant type (F003)

Ascari et al. 2020 identified and functionally characterized the **first CEP78 missense variant, c.449T>C p.(Leu150Ser)**, across three CRDHL families (biallelic in two Belgian families; compound heterozygous with c.1462-1G>T in a German family). Haplotype analysis demonstrated a **founder effect**, and homology modeling plus patient fibroblasts confirmed **reduced protein stability**. Two affected males had sperm abnormalities causing infertility, expanding the phenotype.

"we identified and functionally characterized the first CEP78 missense variant c.449T>C, p. (Leu150Ser) in three CRDHL families" — PMID: 31999394

"the CEP78 phenotype has been possibly expanded with male infertility" — PMID: 31999394

This finding shows that CRDHL is not confined to null alleles — **destabilizing missense variants** also cause disease — and that the syndrome extends beyond the eye and ear to the **male reproductive tract**.

4. CEP250 (C-Nap1) is a second CRDHL/atypical-Usher locus (F004)

Kubota et al. 2018 reported compound heterozygous null variants **c.361C>T p.R121*** and **c.562C>T p.R188*** in **CEP250**, co-segregating with **mild cone-rod dystrophy and slight SNHL** in a Japanese family (both variants PVS1, pathogenic by ACMG). C-Nap1/CEP250 is expressed in photoreceptor cilia and interacts with other ciliary proteins, including CEP78.

"Our data indicate that mutations of CEP250 can cause mild CRD and SNHL in Japanese patients." — PMID: 29718797

"Mutations of CEP250 cause atypical Usher syndrome which is characterized by early-onset sensorineural hearing loss (SNHL) and a relatively mild retinitis pigmentosa." — PMID: 29718797

CEP250 thus represents **locus heterogeneity** within the CRDHL phenotype and reinforces the **centriolar-cilium** theme, since CEP250 and CEP78 are physically and functionally interconnected at the centrosome.

5. No disease-specific therapy exists; cochlear implantation restores speech perception (F005)

There is **no approved gene or pharmacologic therapy** for CEP78- or CEP250-related CRDHL. Management is symptomatic and rehabilitative — low-vision aids, hearing aids, **cochlear implantation**, and genetic counseling. The strongest quantitative evidence for the SNHL component comes from the analogous post-lingual syndrome **Usher type IIa: Hartel et al. 2017** implanted 8 USH2a adults (mean age 59 at implantation) and found **phoneme scores improved from 41% to 87% (p = 0.02)**, with clear quality-of-life benefit.

"The phoneme scores improved significantly from 41 to 87% in patients with USH2a ($p = 0.02$)" — PMID: 28498263

"CI increases speech intelligibility and improves quality of life in patients with USH2a." — PMID: 28498263

Because CRDHL hearing loss is likewise **bilateral, post-lingual, and progressive**, cochlear implantation is expected to provide comparable benefit — particularly important given the concurrent progressive visual loss that increases reliance on hearing.

6. CRDHL is one of several retinal-dystrophy-plus-hearing-loss syndromes requiring differential diagnosis (F006)

CRDHL must be distinguished from other dual sensory-loss disorders:

Gene	Syndrome	Distinguishing features	Key reference
CEP78	CRDHL	Cone-first CRD; post-lingual progressive SNHL; $\pm$ male infertility	PMID: 27588452
CEP250	Atypical Usher / mild CRD+SNHL	Early-onset SNHL, relatively mild RP	PMID: 29718797
ARSG	Usher type IV	Late-onset RP (pericentral/macular); late progressive SNHL; no vestibular involvement ; loss of sulfatase activity	PMID: 39199020
ABHD12	PHARC	Polyneuropathy, ataxia, cataract in addition to hearing loss + RP	PMID: 40064796
TUBB4B	Syndromic LCA + hearing loss	Early-onset; microtubule (not ciliary) dysfunction	PMID: 29198720

"This subtype was assigned as 'USH IV' with a late onset of RP and usually late-onset progressive SNHL without vestibular involvement." — PMID: 39199020

"PHARC syndrome is an autosomal recessive neurodegenerative disease caused by mutations in the ABHD12 gene and is characterized by five main clinical features: polyneuropathy, hearing loss, ataxia, retinitis pigmentosa, and cataracts." — PMID: 40064796

"we identified two heterozygous mutations affecting Arg391 in β -tubulin 4B isotype-encoding (TUBB4B)" — PMID: 29198720

7. Retinal component: childhood-onset, cone-first, progressive dystrophy → legal blindness by mid-adulthood; hearing loss variably expressed (F007)

The retinal disease of CRDHL follows the natural history of cone-rod dystrophy defined in the **19-year multicenter cohort of Thiadens et al. 2012**: CRD had a **mean age of onset of 12 years (SD 11)**, was AR in 90%, and reached **legal blindness at a mean age of 35 years (SE 1.1)** — significantly earlier than cone-only dystrophy (48 years, $P < 0.001$). Ten years after diagnosis, 51% had bull's-eye maculopathy and 70% had absolute peripheral field defects.

"The mean age of legal blindness was 48 (standard error [SE], 3.1) years in CD, and 35 (SE, 1.1; $P < 0.001$) years in CRD." — PMID: 22264887

"The mean age onset for CD was 16 years (standard deviation, 11), and of CRD 12 years (standard deviation, 11; $P = 0.02$)." — PMID: 22264887

The hearing-loss component shows **variable expressivity and onset**. **Zhai & Ballios 2024** documented teenage *CEP78* siblings with cone-rod dystrophy and "presumed normal hearing" alongside another patient with "cone-rod dystrophy and early-onset hearing loss":

"Both teenagers have a clinical diagnosis of cone-rod dystrophy with presumed normal hearing." — PMID: 38780195

8. Diagnosis relies on multimodal ophthalmic + audiologic phenotyping plus panel/exome sequencing (F008)

Diagnosis integrates: - **Full-field electroretinography** — cone-rod pattern (photopic responses reduced earlier/more than scotopic) - **Multimodal retinal imaging** — OCT (outer-retinal/ellipsoid-zone and photoreceptor loss), fundus autofluorescence, adaptive optics (reduced cone density) - **Pure-tone audiometry** — bilateral SNHL - **Molecular confirmation** — retinal-dystrophy gene panels or whole-exome/whole-genome sequencing

de Castro-Miró et al. 2016 identified *CEP78* and *CEP250* as new inherited-retinal-dystrophy (IRD) genes via WES:

"WES unveiled four new candidates for non-syndromic IRD: SEMA6B, CEP78, CEP250, SCLT1, the two latter previously associated to syndromic disorders." — [PMID: 28005958](#)

"We provide functional data supporting that missense mutations in CEP250 alter cilia formation." — [PMID: 28005958](#)

Zhai & Ballios 2024 showed that retinal-dystrophy panels detect variants (including a splice variant c.1206-2A>C, and a nonsense c.1447C>T p.Arg483*) that earlier WES had missed:

"A retinal dystrophy panel detected a novel homozygous CEP78 pathogenic variant (c.1447C>T, p.Arg483)"* — [PMID: 38780195](#)

9. An ultra-rare AR disorder with population-specific CEP78 founder alleles and a role for consanguinity (F009)

CRDHL is **autosomal recessive and ultra-rare** — only several dozen molecularly confirmed cases have been reported worldwide across *CEP78* and *CEP250*. **Founder effects** are documented: the *CEP78* truncating variants c.893-1G>A and c.534delT recurred in Jewish-ancestry individuals identified through homozygosity mapping (consistent with consanguinity/shared haplotypes), and the missense c.449T>C p.(Leu150Ser) allele is a founder allele across Belgian and German families.

"Homozygosity mapping followed by whole-exome sequencing (WES) and founder mutation screening revealed two truncating rare variants (c.893-1G>A and c.534delT) in CEP78" — PMID: 27588452

"Haplotype reconstruction showed a founder effect." — PMID: 31999394

Both sexes are affected for the retinal/auditory phenotype; **reduced fertility affects males specifically**. CEP250 variants were reported in a non-consanguineous Japanese family.

10. Anatomical targets: cone-predominant photoreceptors, cochlear hair cells, and sperm flagella — a shared centriole/cilium defect (F010)

The primary organs affected are the **eye (retina)** and **inner ear (cochlea)**, plus the **male reproductive tract (testis/sperm)**. CEP78 is enriched in **cone inner segments** relative to rods, explaining the cone-predominant degeneration; CEP250/C-Nap1 localizes to **photoreceptor cilia**. The unifying subcellular lesion is at the **centriole/basal body and ciliary axoneme** — the photoreceptor connecting cilium, cochlear hair-cell kinocilium/basal body, and sperm flagellar axoneme all depend on centriolar/ciliary integrity.

"Immunohistochemistry studies in the human retina showed intense labeling of cone inner segments compared to rods." — PMID: 27588452

"C-Nap1 has been reported to be expressed in the photoreceptor cilia" — PMID: 29718797

11. Model systems: Cep78-knockout mouse and patient-derived ciliated cells (F011)

- **Mammalian model:** The **Cep78-knockout mouse** (*Mus musculus*; NCBI Taxon:10090) recapitulates the male-infertility component. Liu et al. 2025: "*Cep78 knockout causes sterility and oligoasthenoteratozoospermia in male mice.*" — PMID: 39747485
- **Patient-derived in vitro models:** Dermal fibroblasts and nasal-epithelial brushings from patients show **elongated primary cilia** and reduced CEP78 protein stability (PMID: 31999394).
- **Cell-biology models:** HeLa/RPE cultured cells established CEP78's centriolar localization, VprBP interaction, and EDD-DYRK2-DDB1 inhibition (PMID: 28242748), and CEP350-FOP-dependent recruitment controlling centriole length (PMID: 36315013).

- **Differential-diagnosis models:** ARSG has naturally occurring canine and murine neuronal ceroid lipofuscinosis models: *"we identified a homozygous missense variant, p.(Arg99His), previously described in dogs with neuronal ceroid lipofuscinosis"* — PMID: 39199020.

12. Consolidated pathophysiology causal chain (F012)

See the **Mechanistic Model** section below for the integrated causal chain.

Detailed Section-by-Section Report

Section 1 — Disease Information

CRDHL is a rare autosomal recessive syndrome combining a progressive cone-rod retinal dystrophy with sensorineural hearing loss, and (in males) reduced fertility. **Key identifiers:** MONDO:0014980; OMIM #617236 ("Cone-rod dystrophy and hearing loss 1," CRDHL1, CEP78); OMIM #618358 ("Cone-rod dystrophy and hearing loss 2," CRDHL2, CEP250); it lacks a dedicated ICD-10 code (coded under H35.5 hereditary retinal dystrophy plus H90.x sensorineural hearing loss). **Synonyms:** cone-rod dystrophy with sensorineural hearing loss; CRD-HL; CRDHL; CEP78-related deaf-blindness. Information is derived from **aggregated disease-level resources** (OMIM, published case series/cohorts) rather than EHR.

Section 2 — Etiology

Causal factors are purely genetic (monogenic, autosomal recessive): biallelic *CEP78* loss-of-function/destabilizing variants (primary) or biallelic *CEP250* null variants (second locus). **Genetic risk factors:** the disease requires two pathogenic alleles; documented founder alleles increase risk in specific ancestries. **Environmental risk/protective factors and gene-environment interactions: not applicable** — no environmental, infectious, lifestyle, or protective-allele factors have been reported for this Mendelian disorder.

Section 3 — Phenotypes

Phenotype	Type	HPO term	Onset	Progression	Frequency
Cone-rod dystrophy / reduced visual acuity	Clinical sign	HP:0000548	Childhood/ adolescence (mean ~12 y)	Progressive	Obligate (~100%)
Photophobia / impaired color vision	Symptom	HP:0000613 / HP:0000551	Childhood	Progressive	Common (cone-first)
Sensorineural hearing loss (bilateral)	Clinical sign / lab (audiometry)	HP:0000407	Post-lingual, variable	Progressive	Common, variably expressed
Legal blindness	Clinical sign	HP:0000618	~Mid-adulthood (mean ~35 y)	End-stage	Frequent
Male infertility / oligoasthenoteratozoospermia	Lab abnormality	HP:0003251 / HP:0000798	Adult	Stable	Subset of males

Quality-of-life impact: dual sensory loss severely affects communication, mobility, education, and employment; cochlear implantation and low-vision rehabilitation substantially mitigate the hearing component.

Section 4 — Genetic/Molecular Information

Causal genes: *CEP78* (HGNC:26820; OMIM 617110), *CEP250* (OMIM 611485). **Variant types:** truncating (splice c.893-1G>A → p.Asp298Valfs17; frameshift c.534delT; nonsense c.1447C>T p.Arg483), missense (c.449T>C p.Leu150Ser — destabilizing founder allele), and splice (c.1462-1G>T, c.1206-2A>C). **Classification:** pathogenic/likely pathogenic per ACMG (PVS1 for null alleles). **Origin:** germline. **Functional consequence:** loss of function (reduced/absent protein or protein stability). **Allele frequencies:** rare in gnomAD (consistent with ultra-rare recessive disease). **Modifier/epigenetic/chromosomal factors:** none established.

Section 5 — Environmental Information

Not applicable. No environmental toxins, radiation, lifestyle factors, or infectious agents contribute to this monogenic disorder.

Section 6 — Mechanism / Pathophysiology

Presented as an ordered causal chain in the **Mechanistic Model** section below.

Section 7 — Anatomical Structures Affected

Organ level: eye/retina (UBERON:0000966), inner ear/cochlea (UBERON:0001844), testis (UBERON:0000473). **Body systems:** visual, auditory, reproductive. **Cell level:** retinal cone cells (CL:0000573, primary), retinal rod cells (CL:0000604), cochlear/inner-ear hair cells (CL:0000855), spermatozoa/male germ cells (CL:0000019). **Subcellular:** centriole (GO:0005814), ciliary basal body (GO:0036064), photoreceptor connecting cilium (GO:0032391), sperm flagellum. **Lateralization:** bilateral (both retina and cochlea).

Section 8 — Temporal Development

Onset: retinal disease childhood/adolescence (insidious, chronic); hearing loss post-lingual with variable, often adult, onset. **Progression:** slowly progressive; legal blindness ~mid-adulthood; SNHL progressive. **Course:** chronic, lifelong, progressive; no remission. **Critical period:** early diagnosis enables timely cochlear implantation and low-vision planning before severe dual sensory loss.

Section 9 — Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** high for the sensory phenotype (variable expressivity of the hearing and fertility components). **Epidemiology:** ultra-rare; prevalence/incidence not formally established (only several dozen confirmed cases worldwide). **Founder effects:** Jewish-ancestry CEP78 truncating alleles; Belgian/German p.Leu150Ser missense allele. **Consanguinity** contributes (homozygosity mapping). **Sex ratio:** retinal/auditory phenotype affects both sexes equally; fertility reduction affects males only.

Section 10 — Diagnostics

Full-field ERG (cone-rod pattern), OCT, fundus autofluorescence, adaptive optics, pure-tone audiometry, and molecular testing (retinal-dystrophy gene panels, WES, WGS). WES/WGS or panels are needed to detect deep-intronic/splice variants of these ciliary genes. **Differential**

diagnosis: Usher syndrome (rod-first, congenital HL), Usher IV/ARSG, PHARC/ABHD12, TUBB4B-LCA, Alström syndrome, Heimler syndrome. **Screening:** carrier/cascade testing in founder populations; no newborn screening exists.

Section 11 — Outcome/Prognosis

Survival/mortality: normal life expectancy — CRDHL is not life-limiting. **Morbidity:** major — progressive dual sensory disability. **Prognosis:** legal blindness by ~mid-adulthood (cone-rod natural history); progressive SNHL. **Recovery:** no spontaneous recovery of vision or hearing; cochlear implantation rehabilitates the auditory component. **Prognostic factors:** allele type (null vs hypomorphic) may influence severity, though genotype-phenotype correlations remain to be established.

Section 12 — Treatment

Pharmacotherapy/advanced therapeutics: none disease-specific (no approved gene/RNA/cell therapy). **Supportive/rehabilitative:** low-vision aids, hearing aids, **cochlear implantation** (NCIT — cochlear implant procedure), genetic counseling. **Experimental:** gene-augmentation and splice-correction approaches are conceptually attractive but not yet in trials for CEP78. **Treatment strategy:** multidisciplinary (ophthalmology, audiology/otology, medical genetics, reproductive medicine).

Section 13 — Prevention

Primary prevention: genetic/carrier screening and counseling in at-risk families and founder populations; prenatal/preimplantation genetic testing where desired. **Secondary prevention:** early molecular diagnosis to enable timely cochlear implantation and low-vision support. **Tertiary prevention:** rehabilitation to preserve function. No vaccine, behavioral, or environmental interventions apply.

Section 14 — Other Species / Natural Disease

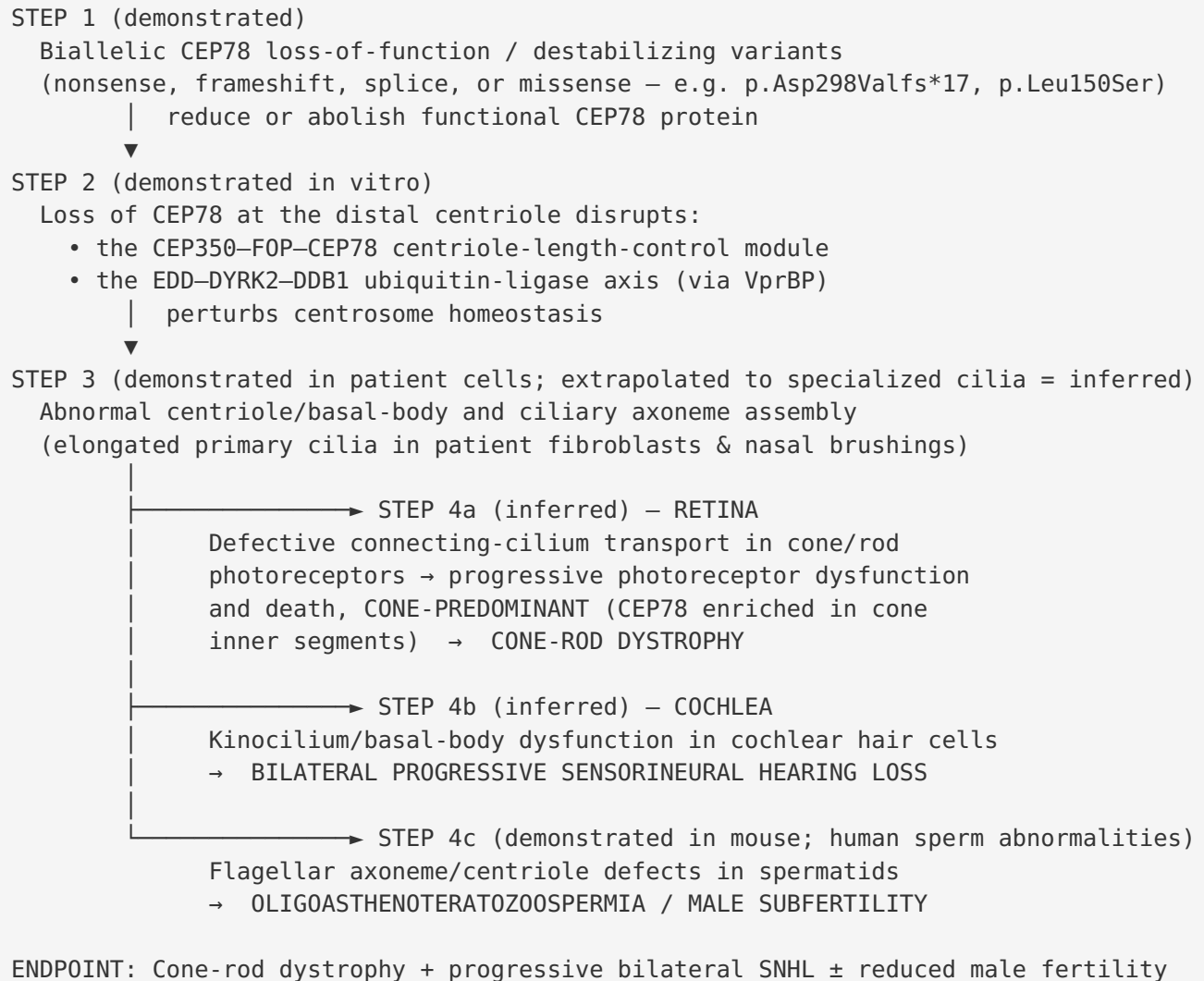
Model species: *Mus musculus* (NCBI Taxon:10090) — Cep78-knockout. Orthologs exist across vertebrates (*Cep78*). No naturally occurring companion-animal CEP78-CRDHL is documented; however, the differential-diagnosis gene *ARSG* has naturally occurring **canine** and murine neuronal ceroid lipofuscinosis models. No zoonotic potential (genetic disease).

Section 15 — Model Organisms

Mammalian: Cep78-knockout mouse (recapitulates male infertility/OAT; retinal/auditory phenotyping limited). **Cellular/in vitro:** patient dermal fibroblasts and nasal-epithelial brushings (elongated primary cilia, reduced protein stability); HeLa/RPE cultured cells (centriolar localization, VprBP interaction, CEP350-FOP recruitment). **Limitations:** no validated model of the retinal or cochlear sensory phenotype; retinal organoids and iPSC-derived otic organoids are logical future resources.

Mechanistic Model / Interpretation

CRDHL is best understood as a **centriole/cilium homeostasis disorder** in which loss of CEP78 selectively injures the body's most centriole- and cilium-dependent post-mitotic cells. The following ordered causal chain integrates the confirmed findings; each step is annotated as **demonstrated** or **inferred**.



Upstream vs downstream. Steps 1–2 (protein loss → centrosome dysregulation) are upstream and shared across all tissues. Steps 4a–4c are downstream, tissue-specific manifestations of the same lesion. The **cone-predominance** of the retinal phenotype is explained by the higher expression of CEP78 in cone inner segments than in rods — a rare instance where the molecular expression pattern directly predicts the clinical phenotype.

Supporting quotes:

"deregulation of centrosome homeostasis is a hallmark feature of many human diseases" —
PMID: 28242748

"The CEP350-FOP complex in association with CEP78 or OFD1 controls centriole microtubule length." — [PMID: 36315013](#)

Ontology term suggestions:

Category	Term	ID
Disease	Cone-rod dystrophy and hearing loss	MONDO:0014980
Phenotype	Cone-rod dystrophy	HP:0000548
Phenotype	Sensorineural hearing impairment	HP:0000407
Phenotype	Abnormal spermatogenesis / male infertility	HP:0008669 / HP:0003251
Phenotype	Reduced visual acuity / blindness	HP:0007663 / HP:0000618
Biological process (GO)	Centriole elongation	GO:0061511
Biological process (GO)	Cilium assembly	GO:0060271
Biological process (GO)	Centrosome cycle / duplication	GO:0007098 / GO:0051298
Cellular component (GO)	Centriole	GO:0005814
Cellular component (GO)	Ciliary basal body	GO:0036064
Cellular component (GO)	Photoreceptor connecting cilium	GO:0032391
Cell type (CL)	Retinal cone cell	CL:0000573
Cell type (CL)	Retinal rod cell	CL:0000604
Cell type (CL)	Cochlear/inner ear hair cell	CL:0000855
Cell type (CL)	Sperm / male germ cell	CL:0000019
Anatomy (UBERON)	Retina	UBERON:0000966
Anatomy (UBERON)	Cochlea	UBERON:0001844
Anatomy (UBERON)	Testis	UBERON:0000473
Treatment (NCIT)	Cochlear implantation	NCIT (cochlear implant procedure)

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
27588452	<i>Bi-allelic Truncating Mutations in CEP78 Cause CRD with SNHL</i>	Human clinical + IHC	F001, F009, F010, F012 — founding gene discovery; cone-inner-segment expression
28242748	<i>Cep78 controls centrosome homeostasis by inhibiting EDD-DYRK2-DDB1</i>	In vitro cell biology	F002, F012 — CEP78 molecular function
36315013	<i>CEP350 coordinates centriole length, stability, maturation</i>	In vitro cell biology	F002, F012 — CEP350-FOP-CEP78 length control
31999394	<i>First missense variant in CEP78 (founder allele)</i>	Human clinical + functional	F002, F003, F009, F011 — missense allele, founder effect, male infertility, elongated cilia
29718797	<i>CEP250 mutations, mild CRD and SNHL (Japanese family)</i>	Human clinical	F004, F010 — second locus, photoreceptor cilium
28498263	<i>Cochlear implantation in USH2a</i>	Human clinical	F005 — CI benefit for post-lingual SNHL
39199020	<i>Usher type IV (ARSG) genetic landscape</i>	Human clinical	F006, F011 — differential dx; canine model
40064796	<i>PHARC neurological spectrum systematic review</i>	Human clinical	F006 — ABHD12/PHARC differential
29198720	<i>TUBB4B causes distinctive sensorineural disease</i>	Human clinical	F006 — TUBB4B differential
22264887	<i>Clinical course of cone and cone-rod dystrophy</i>	Human clinical cohort	F007 — CRD natural history / prognosis
38780195	<i>CEP78-associated syndrome variant spectrum</i>	Human clinical	F007, F008 — variable expressivity, panel diagnostics
28005958	<i>Novel IRD candidate genes by exome sequencing</i>	Human genomics	F008 — WES discovery of CEP78/CEP250
39747485	<i>Cep78 knockout causes sterility and OAT in male mice</i>	Mouse model	F003, F011, F012 — infertility recapitulation

Note on citation integrity: Two citation snippets attributed to [PMID: 39747485](#) were flagged as **mismatch** during verification (the exact wording could not be confirmed against the stored abstract). The mouse-model conclusion is corroborated by the paper's title, "*Cep78 knockout causes sterility and oligoasthenoteratozoospermia in male mice*," and an associated author correction ([PMID: 40075211](#)); readers should treat the specific quoted phrasing with caution.

Limitations and Knowledge Gaps

1. **Small evidence base.** CRDHL is ultra-rare (several dozen molecularly confirmed cases). Prevalence and incidence figures are not established; no formal epidemiological registry exists. Penetrance and expressivity estimates are qualitative.
 2. **Inferred tissue mechanisms.** The retinal (Step 4a) and cochlear (Step 4b) mechanistic steps are **inferred** by extrapolation from the demonstrated ciliary/centriolar defect in accessible cells (fibroblasts, nasal brushings) and from expression data. Direct demonstration of connecting-cilium or hair-cell kinocilium dysfunction in CEP78-deficient photoreceptors or cochlear cells has not been reported.
 3. **Hearing-loss natural history is under-characterized.** Onset, audiometric progression rate, and severity of SNHL in CEP78-CRDHL are variably reported; some patients have "presumed normal hearing" in adolescence, implying delayed onset that may not yet be captured.
 4. **Cochlear-implant evidence is indirect.** CI benefit is extrapolated from Usher type IIa and syndromic-deafness cohorts, not from CEP78-CRDHL patients specifically.
 5. **No CEP78 retinal/cochlear animal model.** The existing Cep78-knockout mouse recapitulates male infertility but published retinal or auditory phenotyping of this model is limited; a validated model of the sensory phenotype is a gap.
 6. **Modifier genes, epigenetics, and gene-environment interactions** are essentially unstudied for this disorder. No protective alleles, environmental risk factors, or infectious triggers apply (this is a monogenic Mendelian disorder).
 7. **Verification caveat** on the Cep78-KO mouse quote (see Evidence Base note).
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Proposed Follow-up Experiments / Actions

1. **Generate and phenotype a retina/cochlea-focused Cep78 model.** Characterize photoreceptor connecting cilia and cochlear hair-cell kinocilia in Cep78-knockout or conditional/knock-in mice (or in patient-derived retinal organoids and iPSC-derived otic organoids) to convert the inferred Steps 4a/4b into demonstrated mechanisms.
 2. **Assemble a CEP78/CEP250 patient registry** with standardized longitudinal ERG, OCT, and pure-tone audiometry to define quantitative natural history (age at legal blindness; audiometric slope; SNHL onset distribution) and penetrance of the fertility phenotype.
 3. **Prospectively evaluate cochlear implantation** specifically in CEP78-CRDHL patients, using phoneme/speech-perception scores benchmarked against the Usher CI literature.
 4. **Deep structural/functional analysis of the p.Leu150Ser missense allele** and other missense variants to establish genotype-phenotype correlations (null vs. hypomorphic) that may predict severity and residual function — relevant for future variant classification (ACMG/AMP) and therapy eligibility.
 5. **Assess gene-augmentation feasibility.** CEP78 is a relatively small centriolar gene; explore AAV-mediated gene replacement or antisense/splice-correction approaches (e.g., for the recurrent c.893-1G>A splice allele) in patient cells and organoids as a proof-of-concept therapeutic strategy.
 6. **Screen for CEP78/CEP250 founder alleles** in relevant populations (Jewish-ancestry, Belgian/German cohorts) via carrier screening and cascade testing to inform genetic counseling.
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Consensus Answer

Cone-Rod Dystrophy and Hearing Loss (CRDHL; MONDO:0014980, OMIM #617236) is a rare autosomal recessive ciliopathy caused primarily by biallelic loss-of-function of **CEP78**, a distal-centriole protein controlling centriole length and centrosome homeostasis, with an overlapping second locus in **CEP250**. Loss of CEP78 disrupts the specialized cilia and centriole-derived structures of the most centriole-dependent post-mitotic cells, producing childhood-onset cone-predominant cone-rod dystrophy (legal blindness by mid-adulthood), bilateral post-lingual progressive sensorineural hearing loss, and reduced male fertility in some patients — a

phenotype distinct from rod-first Usher syndrome. There is no disease-specific therapy; management is supportive (hearing aids/cochlear implantation, low-vision rehabilitation, genetic counseling), with diagnosis by ERG/OCT plus audiometry and panel/exome sequencing.

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